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May 12, 2026

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1 Introduction

Urban landscapes are often warmer, on average, than their surrounding rural counterparts. This phenomenon (known as the Urban Heat Island Effect, and hereafter referred to as the UHI) is largely driven by the lack of vegetated surfaces (Yang et al., 2016). The primary surface materials in urban landscapes (e.g. concrete, tar, asphalt and gravel) absorb more solar radiation (albedos typically ranging from 0.1 to 0.2) than more natural landscapes with very little anthropogenic activity (albedos typically ranging from 0.7 to 0.85) Owing to the UHI among other factors, climate-change-induced global warming is projected to disproportionately affect urban centers (CITATION) ; citites have predicted temerature increases of X degrees C, an X degree increase from the global average (CITATION). [need to talk about why heating matters or no?].

Faced with these unique risks, many municipalities around the world (e.g. Basel, Switzerland and Berlin, Germany) have adoped green roof by-laws (hereafter referred to as GRBLs) to mitigate against the UHI. Green roofs are defined by Berardi et al. (2014) as “roofs with plants in their final layer”, and offer many building and/or city-level services including stormwater management and increased public access to nearby green spaces (Berardi et al., 2014). Most obviously, green roofs decrease the overall solar absorbance of a municipality by increasing the vegetated area, theoretically reducing the UHI.

Toronto is the first city in North America to have had a By-law requiring the construction of green roofs (City of Toronto, 2026). From January 2010 to November 2025, all new developments and extensions with a floor area in excess of 2000 m² were required to construct a green roof, defined by the City as “an extension of [a roof], built on top of a human-made structure, that allows vegetation to grow in a growing medium” (City of Toronto, 2026). Complimenting their

*Code and data are available at: https://github.com/bennypk1/starter_folder.

GRBL, the City also implemented an Eco-Roof Incentive Program (ERIP) to support the voluntary installation of green roofs on homes and buildings (City of Toronto, 2026).

Recent research has found that, while overall area is the most important factor in determining a green roofs’s ability to offset the UHI, the relative spatial arrangement of green spaces in a city matters as well, where cities with (Berardi et al., 2014). In this report, we sought to assess whether or not Toronto’s GRBL was effective in homogenizing the distribution of vegetated area in the city, effectively optimizing the city’s UHI offset. We leveraged data from Open Data Toronto to (1) document the spatial distribution of current ground-level “green spaces” (e.g. parks, etc.), and (2) document the spatial distribution of green roofs, and (3) analyze whether or not green roofs “fill the gaps” missed on the surface. We hypothesize that increased rates of development in downtown areas (and therefore higher absolute green roof numbers) would lead to

[Findings]

The remainder of this report is structured as follows. Section 2 provides an overview of the datasets used, as well as summaries of variables relevant for analysis. Section 3 details our findings [and will primarily consist of comparative maps], Section 4 discussion and limitations.

2 Data

Two datasets were used in this report, both sourced from the City of Toronto’s Open Data website (CITATION), accessed on May 11, 2026. These are: “Building Permits - Green Roofs” (CITATION), and “Green Spces” (CITATION), updated weekly and monthly respectively. [More information about the specific datasets is available below]. All data acquisition, cleaning, tabulation, and visualization tasks were conducted within the R Statistical Software (R Core Team, 2023). Notable packages used in this report include `opendatatoronto` (Gelfand, 2022), `tidyverse` (Wickham et al., 2019), `canadianmaps` (Cayen, 2024), [add rest of libraries used here].

2.1 Building Permits - Green Roofs

The “Building Permits - Green Roofs” dataset consists of 1234 building permits related to Toronto’s GRBL (n=1193) and ERIP (n=41). Both the GRBL and ERIP were adopted in 2009 by Toronto City Council, and entered into effect on January 31, 2010. The GRBL was eventually repealed in November 2025, making green roofs voluntary for private developers (City of Toronto, 2026). Permit information (described in more detail below) of all structures for which the GRBL or ERIP has applied are made public and updated weekly on the City’s Open Data website (City of Toronto, 2026), accessed for this report on May 6, 2026. The raw dataset also contained an additional 15 definitively cancelled projects and 5 duplicate entries, which were removed during cleaning.

Each permit is assigned a Permit Number, which consists of the last two digits of its application year (e.g. 11 for 2011), followed by a 6-digit IBMS-generated sequence. After issuance, permit applications may be revised, the proposed changes of which may qualify for GRBL or ERIP. Such revisions are recorded using the same Permit Number, and distinguished from the original permit application by an integer value (called the Revision Number). Therefore, identifying truly unique green roofs requires specifying both a Permit and Revision Number. Permit-revision pairs are each assigned types (e.g. “New Building”, “Small Residential Projects”), a current status (e.g. “Closed”, “Revision Issued”), and a written description of the proposed project. Dates of permit application, issuance, and completion are also provided where applicable. In addition, the full address, forward sortation area (CFSAUID ; i.e. the first three digits of the postal code), type (e.g. “University”, “SFD – Detached”), and green roof area (in m²) of the permit’s associated structure is provided.

For this analysis, our variables of interest from this dataset are CFSAUID and green roof area. Table 1 demonstrates what the dataset looked like after cleaning.

Table 1: Table 1: First 4 Rows of Cleaned Green Roof Data

ID	G_AREA	CFSAUID
03 141785_1	18.36	M8Y
09 176602_1	375.00	M5S
10 115481_0	615.00	M5A
10 120075_0	1470.00	M5B

2.2 Green Spaces Dataset

The “Green Spaces” dataset represents a collection of 3372 geospatial representations of parks, beaches, ravines, etc. in Toronto, and is jointly maintained by the Geospatial Competency Centre and the Parks, Forestry & Recreation division (CITATION). Importantly, this dataset does not include green roofs. The R packages `sf`, `geojsonsf`, and `canadianmaps` were used to parse these representations to identify their area and assign them each an FSA based on their centroid coordinates.

Table 2: Table 2: First 4 Rows of Cleaned Green Spaces Data

ID	G_AREA	CFSAUID
1	1222.710	M1V
2	1517.757	M1B
3	180759.352	M2H
4	1252.153	M1W

2.3 Merging Data by CFSAUID

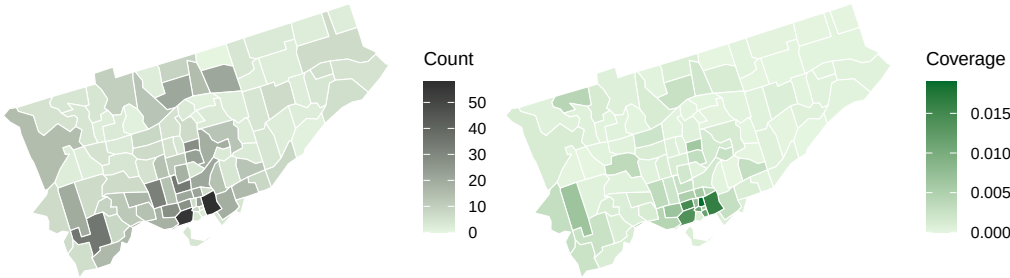
Once parsed, both dataset were joined together by FSA, allowing for effective comparisons of spatial distributions between green spaces and green roofs. 17% (n = 214) of green roof building permits did not have associated FSAs in the original dataset, and thus were necessarily omitted at this stage. To control for the large variance in sizes among FSAs, we defined a new variable, coverage, which is calculated as the proportion of an FSA’s total area that is taken up by green roofs (or green spaces). FSA total area was parsed from data available in the `canadainmaps` package, following a similar procedure to Green Spaces. This allows us to better compare how “green” FSAs are.

Table 3: Table 3: First 4 Rows of Merged Spatial Summary Data

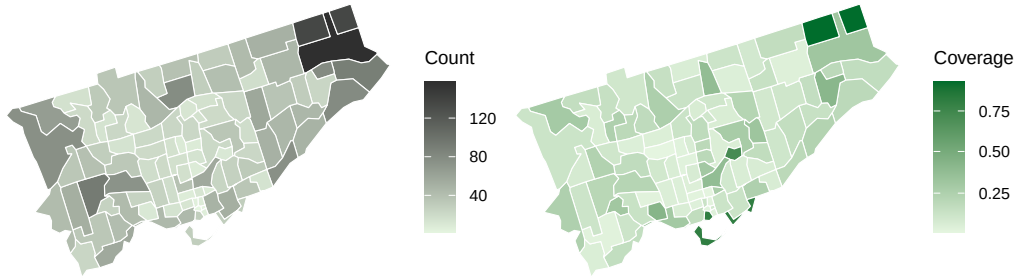
CF- SAUID	geome- try	GREEN_ROOFS_COUNT	GREEN_SPACES_COUNT	GREEN_ROOFS_AREA	GREEN_SPACES_AREA	GREEN_ROOF_COVERAGE	GREEN_SPACE_COVERAGE
M1B	list(list(...	7	158	3778.81	8169009	0.0001481	0.3201125
M1C	list(list(...	5	90	5673.10	2538517	0.0003865	0.1729499
M1E	list(list(...	6	80	4424.10	3162343	0.0002761	0.1973799
M1G	list(list(...	4	24	3109.75	3379795	0.0003851	0.4185126

3 Results

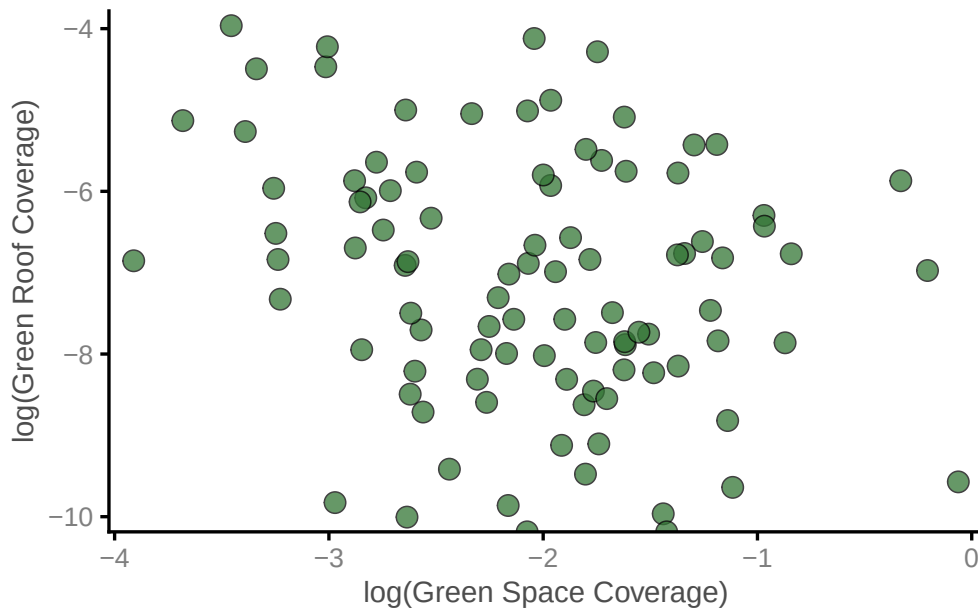
3.1 Visualizing the location and coverage of Toronto’s Green Roofs and Green Spaces



BLllllllll



3.2 Comparing the Distributions of Green Roofs and Green Spaces



In Figure 3 we clearly observe a negative association between green roof and green space coverage, supporting our initial hypothesis.

4 Discussion

[Figure 4: Distribution of Green Roof Areas in the Dataset: should see that a few buildings govern area gained. Should show mean and median to illustrate skew.]

Figure 4 shows that a few structures govern the green roof area in Toronto. Examining these, we see that they are primarily industrial buildings outside of the city centre (CHECK THIS).

Since the GRBL. Although the ERIP locations represent optional adaptations, GRBLs are larger on average, and there are only $n = 41$ ERIPs with a total area of $17,911 \text{ m}^2$ representing only 0.17% of the $1,066,1611 \text{ m}^2$ of total green roof area.

Limitations: - Green roof area is weird, sometimes zero for not reason (get the proportion), and for no clear reason given the original variables provided in the dataset (e.g. no structures systematically, etc.) - Wong et al., 2013 found that many of the green roofs were shaded for a lot of the year, limiting their effects. This logically also applies to parks, so a more thorough investigation requires looking at the 3D geometry of these sites.

Next Steps: - Address Wong et al. - Work has already been done to assess the positive impacts
- Cross-disciplinary studies to identify whether another GRBL is required, and what

Appendix

5 References